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Semiconductor device

Abstract

A semiconductor device includes an insulating support member, a first and a second conductive layer, a first semiconductor element, a first lead, a first detection conductor and a first gate conductor. The first and second conductive layers are disposed on a front surface of the insulating support member. The first semiconductor includes a first and a second electrode on the same side, and a third electrode disposed on the other side and electrically connected to the first conductive layer. The first lead is connected to the first and second conductive layer. The first detection conductor is connected to the first electrode. The first gate conductor is connected to the second electrode. At least one of the first detection conductor and the first gate conductor has an end connected to the first semiconductor element. The end has a coefficient of linear expansion smaller than that of the first conductive layer.

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Background/Summary

CROSS-REFERENCE (1) This application is a continuation application of U.S. application Ser. No. 17/273,532, filed Mar. 4, 2021, which is a national stage of international application PCT/JP2019/035459, filed Sep. 10, 2019, which claims priority to Japanese application 2018-170502, filed Sep. 12, 2018, all of which are incorporated herein by reference, including the original claims.

TECHNICAL FIELD
(1) The present disclosure relates to a semiconductor device provided with a semiconductor element, and more specifically, to a semiconductor device provided with a switching element as the semiconductor element.
BACKGROUND ART
(2) Semiconductor devices with switching elements such as MOSFETs or IGBTs are conventionally known. An example of a semiconductor device that uses a MOSFET is disclosed in Patent Document 1. In the semiconductor device, a semiconductor element is bonded to a lead that constitutes a drain terminal. The semiconductor device has a metal piece connected to a source pad of the semiconductor element and to a lead that constitutes a source terminal. The metal piece, which is made of aluminum, allows for flowing a large amount of current through the semiconductor element. The metal piece also promotes heat dissipation from the semiconductor element, thereby reducing ON-resistance.
(3) The inventor conducted a ΔT.sub.j power cycle test to a device having a configuration similar to the semiconductor device disclosed in Patent Document 1, to find that cracking can occur in the bonding layer (such as solder) interposed between the source pad and the metal piece. This is due to the thermal stress generated because the coefficient of linear expansion of the metal piece is large as compared with the semiconductor element. Such cracking in the bonding layer can be prevented by changing the material for the metal piece to copper (of which coefficient of linear expansion is smaller than that of aluminum). However, conducting a ΔT.sub.j power cycle test to such a configuration using copper revealed that the gate wire connected to the gate pad of the semiconductor element and a sense wire connected to the source pad can be detached from the semiconductor element due to the concentration of thermal stress.
TECHNICAL REFERENCE
Patent Document
(4) Patent Document 1: JP-A-2008-294384
SUMMARY OF THE INVENTION
Problems to be Solved by the Invention
(5) An object of the present disclosure is to provide a semiconductor device that reduces or eliminates the problems described above (cracking of a bonding layer or detachment of a wire) and

provides a high reliability.

Means for Solving the Problems

(6) The semiconductor device provided according to an aspect of the present disclosure includes: an insulating support member having a front surface; a first and a second conductive layer disposed on the front surface; a first semiconductor element having a first side facing the front surface and a second side facing away from the first side in a thickness direction of the insulating support member, where the first semiconductor element is provided with a first and a second electrode on the second side and a third electrode on the first side, and the third electrode is bonded for electrical connection to the first conductive layer; a first lead connected to the first electrode and the second conductive layer; a first detection conductor connected to the first electrode; and a first gate conductor connected to the second electrode. At least one of the first detection conductor and the first gate conductor has an end connected to the first semiconductor element, where the end has a coefficient of linear expansion smaller than a coefficient of linear expansion of the first conductive layer.

(7) Preferably, each of the first detection conductor and the first gate conductor has a pillow part connected to the first semiconductor element and a wire part connected to the pillow part, and the pillow part has a coefficient of linear expansion smaller than the coefficient of linear expansion of the first conductive layer.

(8) Preferably, the pillow part comprises: a first layer made of an alloy containing iron and nickel; and a pair of second layers made of a metal different from the first layer, and the first layer is disposed between the paired second layers in the thickness direction.

(9) Preferably, the pillow part comprises a first layer made of a semiconductor material and a pair of second layers made of a metal, and the first layer is disposed between the paired second layers in the thickness direction.

(10) Preferably, the first detection conductor comprises a metal piece, the first gate conductor comprises a pillow part connected to the first semiconductor element and a wire part connected to the pillow part, and each of the first detection conductor and the pillow part has a coefficient of linear expansion smaller than the coefficient of linear expansion of the first conductive layer.

(11) Preferably, the first detection conductor comprises: a first layer made of an alloy containing iron and nickel; and a pair of second layers made of a metal different from the first layer, and the first layer is disposed between the paired second layers in the thickness direction.

(12) Preferably, each of the first detection conductor and the first gate conductor comprises a metal piece, and each of the first detection conductor and the first gate conductor has a coefficient of linear expansion smaller than the coefficient of linear expansion of the first conductive layer.

(13) Preferably, the semiconductor device further comprises a first detection wiring layer to which the first detection conductor is connected and a first gate wiring layer to which the first gate conductor is connected, where the first detection wiring layer and the first gate wiring layer overlap with the front surface as viewed along the thickness direction.

(14) Preferably, the first detection wiring layer and the first gate wiring layer are disposed on the front surface.

(15) Preferably, the semiconductor device further comprises an insulating layer disposed on the first conductive layer, wherein the first detection wiring layer and the first gate wiring layer are disposed on the insulating layer.

(16) Preferably, the semiconductor device further comprises: a second semiconductor element provided with a first electrode, a second electrode and a third electrode, where the third electrode is bonded for electrical connection to the second conductive layer; a second lead connected to the first electrode of the second semiconductor element; a second detection conductor connected to the first electrode of the second semiconductor element; and a second gate conductor connected to the second electrode of the second semiconductor element. At least one of the second detection conductor and the second gate conductor has an end connected to the second semiconductor

element, and the end has a coefficient of linear expansion smaller than a coefficient of linear expansion of the second conductive layer.
(17) Preferably, each of the second detection conductor and the second gate conductor comprises a pillow part connected to the second semiconductor element and a wire part connected to the pillow part, and the pillow part has a coefficient of linear expansion smaller than the coefficient of linear expansion of the second conductive layer.
(18) Preferably, the second detection conductor comprises a metal piece, the second gate conductor comprises a pillow part connected to the second semiconductor element and a wire part connected to the pillow part, and each of the second detection conductor and the pillow part has a coefficient of linear expansion smaller than the coefficient of linear expansion of the second conductive layer.
(19) Preferably, each of the second detection conductor and the second gate conductor comprises a metal piece, and each of the second detection conductor and the second gate conductor has a coefficient of linear expansion smaller than the coefficient of linear expansion of the second conductive layer.
(20) Preferably, the semiconductor device further comprises: a second detection wiring layer to which the second detection conductor is connected; and a second gate wiring layer to which the second gate conductor is connected, where the second detection wiring layer and the second gate wiring layer overlap with the front surface as viewed along the thickness direction.
(21) Preferably, the semiconductor device further comprises: a first input terminal electrically connected to the first conductive layer; a second input terminal electrically connected to the second lead; and an output terminal electrically connected to the second conductive layer, where each of the first input terminal and the second input terminal is spaced apart from the output terminal in a direction orthogonal to the thickness direction, and the second lead is connected to the second input terminal.
(22) Preferably, the first input terminal and the second input terminal are spaced apart from each other in the thickness direction, and a part of the second input terminal overlaps with the first input terminal as viewed along the thickness direction.
(23) Other features and advantages of the semiconductor device according to the present disclosure will become apparent from the detailed description given below with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a semiconductor device according to a first embodiment;
(2) FIG. **2** is a plan view of the semiconductor device shown in FIG. **1**;
(3) FIG. **3** is a plan view (seen through a sealing resin) of the semiconductor device shown in FIG. **1**;
(4) FIG. **4** is a plan view of the semiconductor device shown in FIG. **3** as seen through a second input terminal, a plurality of first leads and a plurality of second leads;
(5) FIG. **5** is a bottom view of the semiconductor device shown in FIG. **1**;
(6) FIG. **6** is a right side view of the semiconductor device shown in FIG. **1**;
(7) FIG. **7** is a left side view of the semiconductor device shown in FIG. **1**;
(8) FIG. **8** is a front view of the semiconductor device shown in FIG. **1**;
(9) FIG. **9** is a sectional view taken along line IX-IX in FIG. **3**;
(10) FIG. **10** is a sectional view taken along line X-X in FIG. **3**;
(11) FIG. **11** is a view showing a part (at or near a first semiconductor element) of FIG. **3**;
(12) FIG. **12** is a sectional view taken along line XII-XII in FIG. **11**;
(13) FIG. **13** is a sectional view taken along line XIII-XIII in FIG. **11**;

(14) FIG. **14** is a view showing a part (at or near a second semiconductor element) of FIG. **3**;
(15) FIG. **15** is a sectional view taken along line XV-XV in FIG. **14**;
(16) FIG. **16** is a sectional view taken along line XVI-XVI in FIG. **14**;
(17) FIG. **17** is a sectional view of a part (at or near a first semiconductor element) of a semiconductor device according to a first variation of the first embodiment;
(18) FIG. **18** is a sectional view of a part (at or near a first semiconductor element) of the semiconductor device shown in FIG. **17**;
(19) FIG. **19** is a plan view of a part (at or near a first semiconductor element seen through the sealing resin) of a semiconductor device according to a second variation of the first embodiment;
(20) FIG. **20** is a sectional view taken along line XX-XX in FIG. **19**;
(21) FIG. **21** is a plan view of a part (at or near a second semiconductor element seen through the sealing resin) of the semiconductor device shown in FIG. **19**;
(22) FIG. **22** is a sectional view taken along line XXII-XXII in FIG. **21**;
(23) FIG. **23** is a plan view of a part (at or near a first semiconductor element seen through the sealing resin) of a semiconductor device according to a third variation of the first embodiment;
(24) FIG. **24** is a sectional view taken along line XXIV-XXIV in FIG. **23**;
(25) FIG. **25** is a plan view of a part (at or near a second semiconductor element seen through the sealing resin) of the semiconductor device shown in FIG. **23**;
(26) FIG. **26** is a sectional view taken along line XXVI-XXVI in FIG. **25**;
(27) FIG. **27** is a plan view (seen through the sealing resin) of a semiconductor device according to a second embodiment;
(28) FIG. **28** is a bottom view of the semiconductor device shown in FIG. **27**;
(29) FIG. **29** is a sectional view taken along line XXIX-XXIX in FIG. **27**;
(30) FIG. **30** is a sectional view taken along line XXX-XXX in FIG. **27**;
(31) FIG. **31** is a view showing a part (at or near a first semiconductor element) of FIG. **27**;
(32) FIG. **32** is a view showing a part (at or near a second semiconductor element) of FIG. **27**;
(33) FIG. **33** is a plan view of a part (at or near a first semiconductor element seen through the sealing resin) of a semiconductor device according to a first variation of the second embodiment;
(34) FIG. **34** is a plan view of a part (at or near a second semiconductor element seen through the sealing resin) of the semiconductor device shown in FIG. **33**;
(35) FIG. **35** is a plan view of a part (at or near a first semiconductor element seen through the sealing resin) of a semiconductor device according to a second variation of the second embodiment; and
(36) FIG. **36** is a plan view of a part (at or near a second semiconductor element seen through the sealing resin) of the semiconductor device shown in FIG. **35**.

MODE FOR CARRYING OUT THE INVENTION

(37) Various embodiments and their variations according to the present disclosure are described below based on the drawings.

First Embodiment

(38) A semiconductor device A**10** according to a first embodiment is described below based on FIGS. **1-16**.
(39) As shown in FIGS. **3**, **4**, **9** and **10**, the semiconductor device A**10** includes an insulating support member (insulating substrate) **10**. In the illustrated example, the insulating support member **10** is made up of two substrates, i.e., a first substrate **10**A and a second substrate **10**B, but the present disclosure is not limited to such a configuration. The semiconductor device A**10** also includes a first conductive layer **20**A, a second conductive layer **20**B, a first detection wiring layer **21**A, a first gate wiring layer **22**A, a plurality of first semiconductor elements **40**A, a plurality of first leads **51**A, a plurality of first detection conductors **52**A and a plurality of first gate conductors **53**A. In addition to these, the semiconductor device A**10** includes a second detection wiring layer **21**B, a second gate wiring layer **22**B, a first input terminal **31** (see FIGS. **4** and **10**), a second input

terminal **32** (see FIGS. **3** and **10**), an output terminal **33** (see FIGS. **3**, **4** and **10**), a pair of detection terminals **34** (see FIGS. **3** and **4**), a pair of gate terminals **35**, a plurality of dummy terminals **36**, a plurality of second semiconductor elements **40**B, a plurality of second leads **51**B, a plurality of second detection conductors **52**B, a plurality of second gate conductors **53**B, a sealing resin **60** and a metal substrate **69** (see FIGS. **9** and **10**). In the illustrated example, the metal substrate **69** includes two regions corresponding to the first substrate **10**A and the second substrate **10**B, respectively, but the present disclosure is not limited to such a configuration. The semiconductor device A**10** is a power converter (power module) in which the first semiconductor elements **40**A and the second semiconductor elements **40**B are MOSFETS, for example. The semiconductor device A**10** may be used for a driving source of a motor, an inverter device for various electric appliances, and a DC/DC converter, for example. In FIG. **3**, the sealing resin **60** is illustrated as transparent for convenience of understanding. In FIG. **4**, the second input terminal **32**, the first leads **51**A and the second leads **51**B, which are shown in FIG. **3**, are also illustrated as transparent for convenience of understanding. In these figures, the sealing resin **60**, the second input terminals **32**, the first leads **51**A and the second leads **51**B are indicated by imaginary lines (two-dot chain lines).
(40) In the description of the semiconductor device A**10**, the thickness direction of the insulating support member **10** is referred to as “thickness direction z”. A direction orthogonal to the thickness direction z is referred to as “first direction x”. The direction orthogonal to both the thickness direction z and the first direction x is referred to as “second direction y”. As shown in FIGS. **1** and **2**, the semiconductor device A**10** is rectangular as viewed along the thickness direction z, i.e., as viewed in plan. The first direction x corresponds to the longitudinal direction of the semiconductor device A**10**. The second direction y corresponds to the widthwise direction of the semiconductor device A**10**. Also, in the description of the semiconductor device A**10**, for convenience of understanding, the side in the first direction x on which the first input terminal **31** and the second input terminal **32** are located is referred to as “first side in the first direction”. The side in the first direction x on which the output terminal **33** is located is referred to as “second side in the first direction x”. Note that the terms “thickness direction z”, “first direction x”, “second direction y”, “first side in the first direction x” and “second side in the first direction” are applied to the description of a semiconductor device A**20** given later.
(41) As shown in FIGS. **3**, **9** and **10**, the first conductive layer **20**A, the second conductive layer **20**B and the metal substrate **69** are arranged on the insulating support member **10**. The insulating support member **10** is electrically insulating. The insulating support member **10** is made of a ceramic material that has a high thermal conductivity. Aluminum nitride (AlN) is an example of such a ceramic material.
(42) As shown in FIGS. **3**, **9** and **10**, in the semiconductor device A**10**, the insulating support member **10** includes two substrates, i.e., the first substrate **10**A and the second substrate **10**B. The first substrate **10**A and the second substrate **10**B are spaced apart from each other in the first direction x. The first substrate **10**A is offset toward the first side in the first direction x. The second substrate **10**B is offset toward the second side in the first direction x. As viewed along the thickness direction z, each of the first substrate **10**A and the second substrate **10**B has a rectangular shape with its longer sides extending along the second direction y. Note that the configuration of the insulating support member **10** is not limited to this and may be constituted of a single substrate.
(43) As shown in FIGS. **9** and **10**, each of the first substrate **10**A and the second substrate **10**B has a front surface **101** and a back surface **102**. The front surface **101** faces the side in the thickness direction z on which the first conductive layer **20**A and the second conductive layer **20**B are arranged. The back surface **102** faces away from the front surface **101** in the thickness direction z.
(44) As shown in FIGS. **3**, **9** and **10**, the first conductive layer **20**A is arranged on the front surface **101** of the first substrate **10**A (insulating support member **10**). Along with the second conductive layer **20**B, the first input terminal **31**, the second input terminal **32** and the output terminal **33**, the

first conductive layer **20**A forms a conduction path connecting the semiconductor elements **40**A and the second semiconductor elements **40**B to the outside of the semiconductor device A**10**. The first conductive layer **20**A is made of a metal foil made of copper (Cu) or a copper alloy, for example. As viewed along the thickness direction z, the first conductive layer **20**A has a rectangular shape with its longer sides extending along the second direction y. In the example of the semiconductor device A**10**, the first conductive layer **20**A is formed of a single region, but in another example, the first conductive layer may be divided into a plurality of regions. The number of the regions and shape of the first conductive layer **20**A can be set freely. Note that the surface of the first conductive layer **20**A may be plated with silver (Ag).

(45) As shown in FIGS. **3**, **9** and **10**, the first detection wiring layer **21**A is arranged on the front surface **101** of the first substrate **10**A. Thus, as viewed along the thickness direction z, the first detection wiring layer **21**A overlaps with the front surface **101**. The first detection wiring layer **21**A is offset toward the second side in the first direction x from the first conductive layer **20**A. The first detection wiring layer **21**A is in the form of a strip elongated in the second direction y. The first detection wiring layer **21**A may be made of the same metal foil as the first conductive layer **20**A, for example. Note that the surface of the first detection wiring layer **21**A may be plated with silver.

(46) As shown in FIGS. **3**, **9** and **10**, the first gate wiring layer **22**A is arranged on the front surface **101** of the first substrate **10**A. Thus, as viewed along the thickness direction z, the first gate wiring layer **22**A overlaps with the front surface **101**. The first gate wiring layer **22**A is located between the first conductive layer **20**A and the first detection wiring layer **21**A in the first direction x. The first gate wiring layer **22**A is in the form of a strip elongated in the second direction y. The first gate wiring layer **22**A may be made of the same metal foil as the first conductive layer **20**A. Note that the surface of the first gate wiring layer **22**A may be plated with silver.

(47) As shown in FIGS. **3**, **9** and **10**, the second conductive layer **20**B is arranged on the front surface **101** of the second substrate **10**B (insulating support member **10**). The second conductive layer **20**B is made of a metal foil made of copper or a copper alloy, for example. As viewed along the thickness direction z, the second conductive layer **20**B has a rectangular shape with its longer sides extending along the second direction y. In the example of the semiconductor device A**10**, the second conductive layer **20**B is formed of a single region, but in another example, the second conductive layer may be divided into a plurality of regions. The number of the regions and shape of the second conductive layer **20**B can be set freely. Note that the surface of the second conductive layer **20**B may be plated with silver.

(48) As shown in FIGS. **3**, **9** and **10**, the second detection wiring layer **21**B is arranged on the front surface **101** of the second substrate **10**B. Thus, as viewed along the thickness direction z, the second detection wiring layer **21**B overlaps with the front surface **101**. The second detection wiring layer **21**B is offset toward the first side in the first direction x from the second conductive layer **20**B. The second detection wiring layer **21**B is in the form of a strip elongated in the second direction y. The second detection wiring layer **21**B may be made of the same metal foil as the second conductive layer **20**B, for example. Note that the surface of the second detection wiring layer **21**B may be plated with silver.

(49) As shown in FIGS. **3**, **9** and **10**, the second gate wiring layer **22**B is arranged on the front surface **101** of the second substrate **10**B. Thus, as viewed along the thickness direction z, the second gate wiring layer **22**B overlaps with the front surface **101**. The second gate wiring layer **22**B is located between the second conductive layer **20**B and the second detection wiring layer **21**B in the first direction x. The second gate wiring layer **22**B is in the form of a strip elongated in the second direction y. The second gate wiring layer **22**B may be made of the same metal foil as the second conductive layer **20**B. Note that the surface of the second gate wiring layer **22**B may be plated with silver.

(50) As shown in FIGS. **2-6**, the first input terminal **31** and the second input terminal **32** are located on the first side in the first direction x. DC power (voltage), which is the power to be converted, is

input to the first input terminal **31** and the second input terminal **32**. The first input terminal **31** is the positive electrode (P terminal). The second input terminal **32** is the negative electrode (N terminal). As shown in FIG. **10**, the second input terminal **32** is spaced apart from all of the first input terminal **31**, the first conductive layer **20**A and the second conductive layer **20**B in the thickness direction z. The first input terminal **31** and the second input terminal **32** are metal plates. The material for the metal plates is copper or a copper alloy.
(51) As shown in FIG. **4**, the first input terminal **31** has a first connecting part **311** and a first terminal part **312**. In the first input terminal **31**, the boundary between the first connecting part **311** and the first terminal part **312** is a surface extending along the second direction y and the thickness direction z and containing a first side surface **63**A (described later) of the sealing resin **60** located on the first side in the first direction x. The entirety of the first connecting part **311** is covered with the sealing resin **60**. The part of the first connecting part **311** offset toward the second side of in the first direction x is shaped like comb teeth. This comb-teeth-like part is bonded for electrical connection to the surface of the first conductive layer **20**A. Such bonding is performed by solder bonding or ultrasonic bonding, for example. Thus, the first input terminal **31** is electrically connected to the first conductive layer **20**A.
(52) As shown in FIGS. **4** and **5**, the first terminal part **312** extends from the sealing resin **60** toward the first side in the first direction x. As viewed along the thickness direction z, the first terminal part **312** is rectangular. Opposite sides of the first terminal part **312** in the second direction y are covered with the sealing resin **60**. Other portions of the first terminal part **312** are exposed from the sealing resin **60**. With such an arrangement, the first input terminal **31** is supported by both of the first conductive layer **20**A and the sealing resin **60**.
(53) As shown in FIG. **3**, the second input terminal **32** has a second connecting part **321** and a second terminal part **322**. As viewed along the thickness direction z, the boundary between the second connecting part **321** and the second terminal part **322** of the second input terminal **32** corresponds to the boundary between the first connecting part **311** and the first terminal part **312** of the first input terminal **31**. The second connecting part **321** is in the form of a strip elongated in the second direction y.
(54) As shown in FIGS. **2** and **3**, the second terminal part **322** extends from the sealing resin **60** toward the first side in the first direction x. As viewed along the thickness direction z, the second terminal part **322** is rectangular. Opposite sides of the second terminal part **322** in the second direction y are covered with the sealing resin **60**. Other portions of the second terminal part **322** are exposed from the sealing resin **60**. As shown in FIGS. **3** and **4**, as viewed along the thickness direction z, the second terminal part **322** overlaps with the first terminal part **312** of the first input terminal **31**. As shown in FIG. **10**, the second terminal part **322** is offset from the first terminal part **312** in the sense of the thickness direction z in which the front surface **101** of the insulating support member **10** faces. Note that in the example of the semiconductor device A**10**, the shape of the second terminal part **322** is the same as that of the first terminal part **312**.
(55) As shown in FIGS. **6** and **10**, an insulator **39** is interposed between the first terminal part **312** of the first input terminal **31** and the second terminal part **322** of the second input terminal **32** in the thickness direction z. The insulator **39** is a flat plate. The insulator **39** is electrically insulating and made of insulating paper, for example. As viewed along the thickness direction z, the entirety of the first input terminal **31** overlaps with the insulator **39**. In the second input terminal **32**, part of the second connecting part **321** and the entirety of the second terminal part **322** overlap with the insulator **39**, as viewed along the thickness direction z. These portions overlapping with the insulator **39** as viewed along the thickness direction z are in contact with the insulator **39**. The insulator **39** insulates the first input terminal **31** and the second input terminal **32** from each other. Parts of the insulator **39** (parts on the second side in the first direction x and opposite sides in the second direction y) are covered with the sealing resin **60**.
(56) As shown in FIGS. **3**, **4** and **10**, the insulator **39** includes an interposed part **391** and an

extension **392**. The interposed part **391** is located between the first terminal part **312** of the first input terminal **31** and the second terminal part **322** of the second input terminal **32** in the thickness direction z. The entirety of the interposed part **391** is sandwiched between the first terminal part **312** and the second terminal part **322**. The extension **392** extends from the interposed part **391** toward the first side in the first direction x beyond the first terminal part **312** and the second terminal part **322**. Thus, the extension **392** is offset from the first terminal part **312** and the second terminal part **322** toward the first side in the first direction x. Opposite sides of the extension **392** in the second direction y are covered with the sealing resin **60**.

(57) As shown in FIGS. **2-7** (excluding FIG. **6**), the output terminal **33** is located on the second side in the first direction x. The AC power (voltage) obtained by power conversion by the first semiconductor elements **40**A and the second semiconductor elements **40**B is outputted from the output terminal **33**. The output terminal **33** is a metal plate. The material for the metal plate is copper or a copper alloy. The output terminal **33** has a connecting part **331** and a terminal part **332**. The boundary between the connecting part **331** and the terminal part **332** is a surface extending along the second direction y and the thickness direction z and containing a first side surface **63**A (described later) of the sealing resin **60** located on the second side in the first direction x. The entirety of the connecting part **331** is covered with the sealing resin **60**. The connecting part **331** is provided with a comb-teeth portion **331**A on the first side in the first direction x. The comb-teeth portion **331**A is bonded for electrical connection to the surface of the second conductive layer **20**B. Such bonding is performed by solder bonding or ultrasonic bonding, for example. Thus, the output terminal **33** is electrically connected to the second conductive layer **20**B. As shown in FIGS. **2-5**, the terminal part **332** extends from the sealing resin **60** toward the second side in the first direction x. As viewed along the thickness direction z, the terminal part **332** is rectangular. Opposite sides of the terminal part **332** in the second direction y are covered with the sealing resin **60**. Other portions of the terminal part **332** are exposed from the sealing resin **60**. With such an arrangement, the output terminal **33** is supported by both of the second conductive layer **20**B and the sealing resin **60**.

(58) As shown in FIGS. **3**, **9** and **10**, the first semiconductor elements **40**A are bonded for electrical connection to the first conductive layer **20**A. The first semiconductor elements **40**A are arranged at predetermined intervals along the second direction y. The first semiconductor elements **40**A form an upper arm circuit of the semiconductor device A**10**. Also, as shown in FIGS. **3**, **9** and **10**, the second semiconductor elements **40**B are bonded for electrical connection to the second conductive layer **20**B. The second semiconductor elements **40**B are arranged at predetermined intervals along the second direction y. The second semiconductor elements **40**B form a lower arm circuit of the semiconductor device A**10**. The first semiconductor elements **40**A and the second semiconductor elements **40**B are in staggered arrangement along the second direction y. In the illustrated example of the semiconductor device A**10**, the semiconductor device A**10** includes four first semiconductor elements **40**A and four second semiconductor element **40**B. The number of the first semiconductor elements **40**A and second semiconductor elements **40**B is not limited to this and can be varied according to the performance required for the semiconductor device A**10**.

(59) The first semiconductor elements **40**A and the second semiconductor elements **40**B are the same semiconductor elements. The semiconductor elements may be, for example, MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistor) made by using a semiconductor material mainly composed of silicon carbide (SiC). Note that the first semiconductor elements **40**A and the second semiconductor elements **40**B are not limited to MOSFETs and may be field-effect transistors including MISFETs (Metal-Insulator-Semiconductor Field-Effect Transistor) or bipolar transistors such as IGBTs (Insulated Gate Bipolar Transistor). In the description of semiconductor device A**10**, it is assumed that the first semiconductor elements **40**A and the second semiconductor elements **40**B are n-channel MOSFETs.

(60) As shown in FIGS. **11** and **14**, each of the first semiconductor elements **40**A and the second

semiconductor elements **40**B is rectangular as viewed along the thickness direction z (square in the semiconductor device A**10**). As shown in FIGS. **11-16**, each of the first semiconductor elements **40**A and the second semiconductor elements **40**B includes an element front surface **401**, an element back surface **402**, a first electrode **41**, a second electrode **42**, a third electrode **43** and an insulating film **44**. The element front surface **401** and the element back surface **402** face away from each other in the thickness direction z. Of these, the element front surface **401** faces the side that the front surface **101** of the insulating support member **10** faces.

(61) As shown in FIGS. **11-16**, the first electrode **41** is on the element front surface **401**, i.e., on the side that the front surface **101** of the insulating support member **10** faces in the thickness direction z. A source current flows from inside the first semiconductor element **40**A or the second semiconductor element **40**B to the first electrode **41**.

(62) As shown in FIGS. **11**, **13**, **14** and **16**, the second electrode **42** is on the element front surface **401**, i.e., on the side that the front surface **101** of the insulating support member **10** faces in the thickness direction z. A gate voltage for driving the first semiconductor element **40**A or the second semiconductor element **40**B is applied to the second electrode **42**. The size of the second electrode **42** is smaller than that of the first electrode **41**. In each of the first semiconductor elements **40**A, the second electrode **42** is offset toward one side in the second direction y (the side on which the pair of detection terminals **34**, the pair of gate terminals **35** and the dummy terminals **36** are located). In each of the second semiconductor elements **40**B, the second electrode **42** is offset toward the other side in the second direction y.

(63) As shown in FIGS. **12**, **13**, **15** and **16**, the third electrode **43** is on the element back surface **402**, i.e., on the side facing the front surface **101** of the insulating support member **10** in the thickness direction z. The third electrode **43** extends over the entirety of the element back surface **402**. A drain current flows through the third electrode **43** into the first semiconductor element **40**A or the second semiconductor element **40**B. The third electrode **43** of each of the first semiconductor elements **40**A is bonded for electrical connection to the first conductive layer **20**A with a conductive first bonding layer **29**. The first bonding layer **29** is made of a lead-free solder mainly composed of tin (Sn), for example. Thus, the third electrodes **43** of the first semiconductor elements **40**A are electrically connected to the first conductive layer **20**A. Also, the third electrode **43** of each of the second semiconductor elements **40**B is bonded for electrical connection to the second conductive layer **20**B with a first bonding layer **29**. Thus, the third electrodes **43** of the second semiconductor elements **40**B are electrically connected to the second conductive layer **20**B.

(64) As shown in FIGS. **11-16**, the insulating film **44** is on the element front surface **401**. The insulating film **44** is electrically insulating. As viewed along the thickness direction z, the insulating film **44** surrounds each of the first electrode **41** and second electrode **42**. The insulating film **44** may be made up of, for example, a silicon dioxide (SiO.sub.2) layer, a silicon nitride (Si.sub.3 N.sub.4) layer and a polybenzoxazole (PBO) layer laminated in the mentioned order on the element front surface **401**. Note that, in the insulating film **44**, a polyimide layer may be used instead of the polybenzoxazole layer.

(65) As shown in FIGS. **3** and **9**, the first leads **51**A are connected to the first electrodes **41** of the first semiconductor elements **40**A and the second conductive layer **20**B. As viewed along the thickness direction z, each of the first leads **51**A is in the form of a strip elongated in the first direction x. The first leads **51**A are made of copper or a copper alloy. The end of each first lead **51**A on the first side in the first direction x is connected to the first electrode **41** of a respective first semiconductor element **40**A with a conductive second bonding layer **49**. The second bonding layer **49** is made of a lead-free solder mainly composed of tin (Sn) or baked silver, for example. The end of each first lead **51**A on the second side in the first direction x is connected to the second conductive layer **20**B with a first bonding layer **29**. Thus, the first electrodes **41** of the first semiconductor elements **40**A are electrically connected to the second conductive layer **20**B.

(66) As shown in FIGS. **3** and **10**, the second leads **51**B are connected to the first electrodes **41** of

the second semiconductor elements **40**B and the second input terminal **32**. As viewed along the thickness direction z, each of the second leads **51**B is in the form of a strip elongated in the first direction x. The second leads **51**B are made of copper or a copper alloy. The end surface of each second lead **51**B on the first side in the first direction x is directly connected to the second connecting part **321** of the second input terminal **32**. Thus, the second leads **51**B are integral with the second input terminal **32**. The end of each second lead **51**B on the second side in the first direction x is connected to the first electrode **41** of a respective second semiconductor element **40**B with a second bonding layer **49**. Thus, the first electrodes **41** of the second semiconductor elements **40**B are electrically connected to the second input terminal **32**.
(67) As shown in FIGS. **3** and **11**, the first detection conductors **52**A are connected to the first electrodes **41** of the first semiconductor elements **40**A and the first detection wiring layer **21**A. Thus, the first electrodes **41** of the first semiconductor elements **40**A are electrically connected to the first detection wiring layer **21**A. As shown in FIGS. **11** and **12**, each of the first detection conductors **52**A has a pillow part **521** and a wire part **522**.
(68) As shown in FIG. **12**, the pillow part **521** of each first detection conductor **52**A is connected to the first electrode **41** of a respective first semiconductor element **40**A with a second bonding layer **49**. As shown in FIG. **11**, the pillow part **521** is rectangular as viewed along the thickness direction z. The pillow part **521** has a first layer **521**A and a pair of second layers **521**B. The first layer **521**A is made of an alloy containing iron (Fe) and nickel (Ni). Examples of the alloy include invar (Fe-36Ni), super invar (Fe-32Ni-5Co) and Kovar. The paired second layers **521**B are made of a metal. Examples of the metal include copper, a copper alloy, aluminum and an aluminum alloy.
(69) The first layer **521**A is sandwiched between the paired second layers **521**B in the thickness direction z. In this way, the pillow part **521** is a laminate of a plurality of metal layers in the thickness direction z. The ratio of the thickness t**1** of the first layer **521**A and the thickness t**2** of each second layer **521**B may be t**1**: t**2**=8:1, for example. The coefficient of linear expansion of the pillow part **521** having such a configuration is in a range of 0 to 8×10.sup.−6/° C. In contrast, the coefficient of linear expansion of the first conductive layer **20**A is about 16×10.sup.−6/° C. Thus, the coefficient of linear expansion of the pillow part **521** is smaller than that of the first conductive layer **20**A. Note that the first layer **521**A may be made of a semiconductor material. As the semiconductor material, use may be made of silicon (Si), which has a relatively low electrical resistivity. In such a case again, the coefficient of linear expansion of the pillow part **521** is smaller than that of the first conductive layer **20**A.
(70) As shown in FIG. **11**, the wire part **522** of each first detection conductor **52**A is connected to the pillow part **521** of the first detection conductor **52**A and the first detection wiring layer **21**A. The wire part **522** is inclined at an inclination angle α**1***a* with respect to the first direction x. The wire part **522** may be made of aluminum, an aluminum alloy, copper, a copper alloy or a clad material made by some combination of these.
(71) As shown in FIGS. **3** and **14**, the second detection conductors **52**B are connected to the first electrodes **41** of the second semiconductor elements **40**B and the second detection wiring layer **21**B. Thus, the first electrodes **41** of the second semiconductor elements **40**B are electrically connected to the second detection wiring layer **21**B. As shown in FIGS. **14** and **15**, each of the second detection conductors **52**B has a pillow part **521** and a wire part **522**. The pillow part **521** of each second detection conductor **52**B is connected to the first electrode **41** of a respective second semiconductor element **40**B with a second bonding layer **49**. The wire part **522** of each second detection conductor **52**B is connected to the pillow part **521** of the second detection conductor **52**B and the second detection wiring layer **21**B. The wire part **522** of the second detection conductor **52**B is inclined at an inclination angle α**1***b* with respect to the first direction x. Other configurations of the pillow part **521** and the wire part **522** of each second detection conductor **52**B are the same as those of the pillow part **521** and the wire part **522** of each first detection conductor **52**A, so that the description of such configurations is omitted. Note that the coefficient of linear expansion of

the second conductive layer **20**B is generally equal to that of the first conductive layer **20**A. Thus, the coefficient of linear expansion of the pillow part **521** is smaller than that of the second conductive layer **20**B.

(72) As shown in FIGS. **3** and **11**, the first gate conductors **53**A are connected to the second electrodes **42** of the first semiconductor elements **40**A and the first gate wiring layer **22**A. Thus, the second electrodes **42** of the first semiconductor elements **40**A are electrically connected to the first gate wiring layer **22**A. As shown in FIGS. **11** and **13**, each of the first gate conductors **53**A has a pillow part **531** and a wire part **532**.

(73) As shown in FIG. **13**, the pillow part **531** of each first gate conductor **53**A is connected to the second electrode **42** of a respective first semiconductor element **40**A with a second bonding layer **49**. As shown in FIG. **11**, the pillow part **531** is rectangular as viewed along the thickness direction z. The pillow part **531** has a first layer **531**A and a pair of second layers **531**B. The first layer **531**A is made of an alloy containing iron and nickel. Examples of the alloy are the same as those for the first layer **521**A of the pillow part **521** of the first detection conductor **52**A. The paired second layers **531**B are made of a metal. Examples of the metal are the same as those for the second layers **521**B of the pillow part **521** of the first detection conductor **52**A. The first layer **531**A is sandwiched between the paired second layers **531**B in the thickness direction z. In this way, the pillow part **531** is a laminate of a plurality of metal layers in the thickness direction z. The ratio of the thickness t**1** of the first layer **531**A and the thickness t**2** of each second layer **531**B may be t**1**: t**2**=8:1, for example. The coefficient of linear expansion of the pillow part **531** having such a configuration is in a range of 0 to 8×10.sup.−6/° C. In contrast, the coefficient of linear expansion of the second conductive layer **20**B is about 16×10.sup.−6/° C. Thus, the coefficient of linear expansion of the pillow part **531** is smaller than that of the second conductive layer **20**B. Note that the first layer **531**A may be made of a semiconductor material.

(74) The example of the semiconductor material is the same as that for the first layer **521**A of the pillow part **521** of the first detection conductors **52**A. In such a case again, the coefficient of linear expansion of the pillow part **531** is smaller than that of the second conductive layer **20**B.

(75) As shown in FIG. **11**, the wire part **532** of each first gate conductor **53**A is connected to the pillow part **531** of the first gate conductor **53**A and the first gate wiring layer **22**A. The wire part **532** of the first gate conductor **53**A is inclined at an inclination angle α**2***a* with respect to the first direction x. Examples of the material for the wire part **532** are the same as those for the wire part **522** of the first detection conductor **52**A.

(76) As shown in FIGS. **3** and **14**, the second gate conductors **53**B are connected to the second electrodes **42** of the second semiconductor elements **40**B and the second gate wiring layer **22**B. Thus, the second electrodes **42** of the second semiconductor elements **40**B are electrically connected to the second gate wiring layer **22**B. As shown in FIGS. **14** and **16**, each of the second gate conductors **53**B has a pillow part **531** and a wire part **532**. The pillow part **531** of each second detection conductor **53**B is connected to the second electrode **42** of a respective second semiconductor element **40**B with a second bonding layer **49**. The wire part **532** of each second gate conductor **53**B is connected to the pillow part **531** of the second gate conductor **53**B and the second gate wiring layer **22**B. The wire part **532** of the second gate conductor **53**B is inclined at an inclination angle α**2***b* with respect to the first direction x. Other configurations of the pillow part **531** and the wire part **532** of each second gate conductor **53**B are the same as those of the pillow part **531** and the wire part **532** of each first gate conductor **53**A, so that the description of such configurations is omitted. Note that the coefficient of linear expansion of the second conductive layer **20**B is generally equal to that of the first conductive layer **20**A. Thus, the coefficient of linear expansion of the pillow part **531** is smaller than that of the second conductive layer **20**B.

(77) As shown in FIG. **3**, the pair of detection terminals **34**, the pair of gate terminals **35** and the dummy terminals **36** are adjacent to the insulating support member **10** in the second direction y. These terminals are arranged side by side along the first direction x. In the semiconductor device

A**10**, all of the detection terminals **34**, the gate terminals **35** and the dummy terminals **36** are made from a same lead frame.

(78) As shown in FIG. **3**, of the pair of the detection terminals **34**, one is located adjacent to the first substrate **10**A and the other one adjacent to the second substrate **10**B. The voltage (corresponding to the source current) applied to the first electrodes **41** of the first semiconductor elements **40**A or the second semiconductor elements **40**B is detected from each of the detection terminals **34**. Each of the detection terminals **34** has a connecting part **341** and a terminal part **342**. The connecting part **341** is covered with the sealing resin **60**. Thus, the detection terminals **34** are supported by the sealing resin **60**. Note that the surface of the connecting part **341** may be plated with silver, for example. The terminal part **342** is connected to the connecting part **341** and exposed from the sealing resin **60** (see FIG. **8**). The terminal part **342** is L-shaped as viewed along the first direction x.

(79) As shown in FIG. **3**, the paired gate terminals **35** are located adjacent to the paired detection terminal **34** in the first direction x. To each of the gate terminals **35**, a gate voltage for driving the first semiconductor elements **40**A or the second semiconductor element **40**B is applied. Each of the gate terminals **35** has a connecting part **351** and a terminal part **352**. The connecting part **351** is covered with the sealing resin **60**. Thus, the gate terminals **35** are supported by the sealing resin **60**. Note that the surface of the connecting part **351** may be plated with silver, for example. The terminal part **352** is connected to the connecting part **351** and exposed from the sealing resin **60** (see FIG. **8**). The terminal part **352** is L-shaped as viewed along the first direction x.

(80) As shown in FIG. **3**, the dummy terminals **36** are located opposite the detection terminals **34** with respect to the gate terminals **35** in the first direction x. In the example of the semiconductor device A**10**, six dummy terminals **36** are provided. Of these, three dummy terminals **36** are offset toward the first side in the first direction x. The remaining three dummy terminals **36** are offset toward the second side in the first direction x. Note that the number of the dummy terminals is not limited to this. Also, the semiconductor device A**10** may not include the dummy terminal **36**. Each of the dummy terminals **36** has a connecting part **361** and a terminal part **362**. The connecting part **361** is covered with the sealing resin **60**. Thus, the dummy terminals **36** are supported by the sealing resin **60**. Note that the surface of the connecting part **361** may be plated with silver, for example. The terminal part **362** is connected to the connecting part **361** and exposed from the sealing resin **60** (see FIG. **8**). As shown in FIGS. **6** and **7**, the terminal part **362** is L-shaped as viewed along the first direction x. Note that the shape of the terminal parts **342** of the detection terminals **34** and the shape of the terminal parts **352** of the gate terminals **35** are the same as that of the terminal parts **362**.

(81) As shown in FIG. **3**, the semiconductor device A**10** further includes a pair of first wires **54**A and a pair of second wires **54**B. The first wires **54**A and the second wires **54**B are made of aluminum, for example.

(82) As shown in FIG. **3**, the first wires **54**A are connected to the first detection wiring layer **21**A or the second detection wiring layer **21**B and the detection terminals **34**. In the detection terminals **34**, the first wires **54**A are connected to the surfaces of the connecting parts **341**. Thus, one of the detection terminals **34** that is adjacent to the first substrate **10**A is electrically connected to the first electrodes **41** of the first semiconductor elements **40**A, whereas the other one of the detection terminals **34** that is adjacent to the second substrate **10**B is electrically connected to the first electrodes **41** of the second semiconductor elements **40**B.

(83) As shown in FIG. **3**, the second wires **54**B are connected to the first gate wiring layer **22**A or the second gate wiring layer **22**B and the gate terminals **35**. In the gate terminals **35**, the second wires **54**B are connected to the surfaces of the connecting parts **351**. Thus, one of the gate terminals **35** that is adjacent to the first substrate **10**A is electrically connected to the second electrodes **42** of the first semiconductor elements **40**A, whereas the other one of the gate terminals **35** that is adjacent to the second substrate **10**B is electrically connected to the second electrodes **42** of the

second semiconductor elements **40**B.

(84) As shown in FIGS. **9** and **10**, the sealing resin **60** covers the insulating support member **10**, the first conductive layer **20**A, the second conductive layer **20**B, the first semiconductor elements **40**A and the second semiconductor elements **40**B. The sealing resin **60** further covers the first leads **51**A, the second leads **51**B, the first detection conductors **52**A, the second detection conductors **52**B, the first gate conductors **53**A, the second gate conductors **53**B, the first wires **54**A and the second wires **54**B. The sealing resin **60** may be made of black epoxy resin, for example. As shown in FIGS. **2** and **5-8**, the sealing resin **60** has a top surface **61**, a bottom surface **62**, a pair of first side surfaces **63**A, a pair of second side surfaces **63**B, a plurality of third side surfaces **63**C, a plurality of fourth side surfaces **63**D, a plurality of beveled parts **63**E and a plurality of mounting holes **64**.

(85) As shown in FIGS. **9** and **10**, the top surface **61** faces the side that the front surface **101** of the insulating support member **10** faces in the thickness direction z. The bottom surface **62** faces away from the top surface **61** in the thickness direction z. As shown in FIG. **5**, the metal substrate **69** is exposed from the bottom surface **62**. The bottom surface **62** has a frame-like shape surrounding the metal substrate **69**.

(86) As shown in FIGS. **2** and **5-7**, the paired first side surfaces **63**A are connected to both of the top surface **61** and the bottom surface **62** and face in the first direction x. From the first side surface **63**A on the first side in the first direction x, the first terminal part **312** of the first input terminal **31** and the second terminal part **322** of the second input terminal **32** extend toward the first side in the first direction x. From the first side surface **63**A on the second side in the first direction x, the terminal part **332** of the output terminal **33** extends toward the second side in the first direction x. In this way, part of each of the first input terminal **31** and the second input terminal **32** is exposed from the sealing resin **60** on the first side in the first direction x. Also, part of the output terminal **33** is exposed from the sealing resin **60** on the second side in the first direction x.

(87) As shown in FIGS. **2** and **5-8**, the paired second side surfaces **63**B are connected to both of the top surface **61** and the bottom surface **62** and face in the second direction y. From one of the second side surfaces **63**B are exposed the terminal parts **342** of the detection terminals **34**, the terminal parts **352** of the gate terminals **35** and the terminal parts **362** of the dummy terminals **36**.

(88) As shown in FIGS. **2** and **5-7**, the third side surfaces **63**C are connected to both of the top surface **61** and the bottom surface **62** and face in the second direction y. The third side surfaces **63**C include a pair of third side surfaces **63**C located on the first side in the first direction x and a pair of third side surfaces **63**C located on the second side in the first direction x. In each of the first side and the second side in the first direction x, the paired third side surfaces **63**C face each other in the second direction y. Also, in each of the first side and the second side in the first direction x, the paired third side surfaces **63**C are connected to opposite ends of the relevant first side surface **63**A in the second direction y.

(89) As shown in FIGS. **2** and **5-8**, the fourth side surfaces **63**D are connected to both of the top surface **61** and the bottom surface **62** and face in the first direction x. In the first direction x, the fourth side surfaces **63**D are offset from the first side surfaces **63**A toward the outside of the semiconductor device A**10**. The fourth side surfaces **63**D include a pair of fourth side surfaces **63**D located on the first side in the first direction x and a pair of fourth side surfaces **63**D located on the second side in the first direction x. In each of the first side and the second side in the first direction x, the opposite ends of each fourth side surface **63**D in the second direction y are connected to the relevant second side surface **63**B and the relevant third side surface **63**C.

(90) As shown in FIGS. **2** and **5**, each of the beveled parts **63**E is located at the boundary between a first side surface **63**A and a third side surface **63**C. As viewed along the thickness direction z, the beveled parts **63**E are inclined with respect to both of the first direction x and the second direction y.

(91) As shown in FIG. **9**, the mounting holes **64** penetrate the sealing resin **60** from the top surface **61** to the bottom surface **62** in the thickness direction z. The mounting holes **64** are used in

attaching the semiconductor device A**10** to a heat sink (not shown). As shown in FIGS. **2** and **5**, the edge of the mounting holes **64** is circular as viewed along the thickness direction z. The mounting holes **64** are located at the four corners of the sealing resin **60** as viewed along the thickness direction z.

(92) As shown in FIGS. **9** and **10**, the metal substrate **69** is provided on the entirety of the back surface **102** of the insulating support member **10** (the first substrate **10**A and the second substrate **10**B). Accordingly, the metal substrate **69** includes two regions spaced apart from each other in the first direction x. As shown in FIG. **5**, the metal substrate **69** is exposed from the bottom surface **62** of the sealing resin **60**. The metal substrate **69** is made of a metal foil made of copper (Cu) or a copper alloy, for example. The metal substrate **69** is used in attaching the semiconductor device A**10** to a heat sink, along with the mounting holes **64** of the sealing resin **60**.

First Variation of the First Embodiment

(93) A semiconductor device A**11**, which is the first variation of the semiconductor device A**10**, is described below based on FIGS. **17** and **18**. The semiconductor device A**11** differs from the semiconductor device A**10** in configuration of the pillow parts **521** of the first detection conductors **52**A and the second detection conductors **52**B and the pillow parts **531** of the first gate conductors **53**A and the second gate conductors **53**B.

(94) As shown in FIG. **17**, the first layer **521**A of the pillow part **521** of each first detection conductor **52**A has a lower layer portion **521**C, an upper layer portion **521**D and a frame surface **521**E. The lower layer portion **521**C is located on the lower side of the first layer **521**A. The second layer **521**B located at the lower end of the pillow part **521** adjoins the lower layer portion **521**C. The upper layer portion **521**D is connected to the upper end of the lower layer portion **521**C. As viewed along the thickness direction z, the area of the upper layer portion **521**D is larger than that of the lower layer portion **521**C. The second layer **521**B located at the upper end of the pillow part **521** adjoins the upper layer portion **521**D. Thus, as viewed along the thickness direction z, the second layer **521**B adjoining the upper layer portion **521**D has a larger area than the other second layer **521**B adjoining the lower layer portion **521**C. The frame surface **521**E faces the first electrode **41** of the first semiconductor element **40**A. As viewed along the thickness direction z, the frame surface **521**E surrounds the entire periphery of the lower layer portion **521**C. Though the illustration is omitted, the first layer **521**A of the pillow part **521** of each second detection conductor **52**B also has a lower layer portion **521**C, an upper layer portion **521**D and a frame surface **521**E. These have the same configurations as the lower layer portion **521**C, the upper layer portion **521**D and the frame surface **521**E of the pillow part **521** of the first detection conductors **52**A, so that the description is omitted.

(95) As shown in FIG. **18**, the first layer **531**A of the pillow part **531** of each first gate conductor **53**A has a lower layer portion **531**C, an upper layer portion **531**D and a frame surface **531**E. The lower layer portion **531**C is located on the lower side of the first layer **531**A. The second layer **531**B located at the lower end of the pillow part **531** adjoins the lower layer portion **531**C. The upper layer portion **531**D is connected to the upper end of the lower layer portion **531**C. As viewed along the thickness direction z, the area of the upper layer portion **531**D is larger than that of the lower layer portion **531**C. The second layer **531**B located at the upper end of the pillow part **531** adjoins the upper layer portion **531**D. Thus, as viewed along the thickness direction z, the second layer **531**B adjoining the upper layer portion **531**D has a larger area than the other second layer **531**B adjoining the lower layer portion **531**C. The frame surface **531**E faces the second electrode **42** of the first semiconductor element **40**A. As viewed along the thickness direction z, the frame surface **531**E surrounds the entire periphery of the lower layer portion **531**C. Though the illustration is omitted, the first layer **531**A of the pillow part **531** of each second gate conductor **53**B also has a lower layer portion **531**C, an upper layer portion **531**D and a frame surface **531**E. These have the same configurations as the lower layer portion **531**C, the upper layer portion **531**D and the frame surface **531**E of the pillow part **531** of the first gate conductors **53**A, so that the

description is omitted.

Second Variation of the First Embodiment

(96) A semiconductor device A**12**, which is the second variation of the semiconductor device A**10**, is described below based on FIGS. **19-22**. The semiconductor device A**12** differs from the semiconductor device A**10** in configuration of the first detection conductors **52**A and the second detection conductors **52**B.

(97) As shown in FIG. **19**, as viewed along the thickness direction z, each of the first detection conductors **52**A is in the form of a strip elongated in the first direction x. The width B**1***a* of the first detection conductors **52**A is smaller than the width Ba of the first leads **51**A. Each first detection conductor **52**A is made of an elongated metal piece (metal strip). As shown in FIG. **20**, the end of each first detection conductor **52**A on the first side in the first direction x is connected to the first electrode **41** of a respective first semiconductor element **40**A with a second bonding layer **49**. The end of each first detection conductor **52**A on the second side in the first direction x is connected to the first detection wiring layer **21**A with a first bonding layer **29**.

(98) As shown in FIG. **20**, each of the first detection conductors **52**A has a has a first layer **523** and a pair of second layers **524**. The first layer **523** is made of an alloy containing iron and nickel. Examples of the alloy are the same as those for the first layer **521**A of the pillow part **521** of the first detection conductor **52**A. The paired second layers **524** are made of a metal. Examples of the metal are the same as those for the paired second layers **521**B of the pillow part **521** of the first detection conductor **52**A. The first layer **523** is sandwiched between the paired second layers **524** in the thickness direction z. In this way, the first detection conductor **52**A is a laminate of a plurality of metal layers in the thickness direction z. The ratio of the thickness t**3***a* of the first layer **523** at the end on the first side in the first direction x and the thickness t**4***a* of the first layer **523** at the end on the second side in the first direction x may be t**3***a*: t**4***a*=1:2, for example. The coefficient of linear expansion of the first detection conductor **52**A having such a configuration is in a range of 0 to 8×10.sup.−6/° C. In contrast, the coefficient of linear expansion of the first conductive layer **20**A is about 16×10.sup.−6/° C. Thus, the coefficient of linear expansion of the first detection conductors **52**A is smaller than that of the first conductive layer **20**A. The first layer **523** of the first detection conductor **52**A has a transitional surface **523**A. The transitional surface **523**A is a curved surface located at the section where the thickness of the first layer **523** changes from the thickness t**3***a* to the thickness t**4***a*.

(99) As shown in FIG. **21**, as viewed along the thickness direction z, each of the second detection conductors **52**B is in the form of a strip elongated in the first direction x. The width B**1***b* of the second detection conductors **52**B is smaller than the width Bb of the second leads **51**B. Each second detection conductor **52**B is made of a metal piece. As shown in FIG. **22**, the end of each second detection conductor **52**B on the first side in the first direction x is connected to the second detection wiring layer **21**B with a first bonding layer **29**. The end of each second detection conductor **52**B on the second side in the first direction x is connected to the first electrode **41** of a respective second semiconductor element **40**B with a second bonding layer **49**.

(100) As shown in FIG. **22**, each of the second detection conductors **52**B has a has a first layer **523** and a pair of second layers **524**. The ratio of the thickness t**3***b* of the first layer **523** at the end on the first side in the first direction x and the thickness t**4***b* of the first layer **523** at the end on the second side in the first direction x may be t**3***b*: t**4***b*=2:1, for example. Other configurations of the first layer **523** and the second layers **524** of each second detection conductor **52**B are the same as those of the first layer **523** and the second layers **524** of each first detection conductor **52**A, so that the description of such configurations is omitted. Note that the coefficient of linear expansion of the second conductive layer **20**B is generally equal to that of the first conductive layer **20**A. Thus, the coefficient of linear expansion of the second detection conductor **52**B is smaller than that of the second conductive layer **20**B. The first layer **523** of the second detection conductor **52**B has a transitional surface **523**A. The transitional surface **523**A is a curved surface located at the section

where the thickness of the first layer **523** changes from the thickness t**3***b* to the thickness t**4***b*.
Third Variation of the First Embodiment
(101) A semiconductor device A**13**, which is the third variation of the semiconductor device A**10**, is described below based on FIGS. **23-26**. The semiconductor device A**13** differs from the semiconductor device A**10** in configuration of the first detection conductors **52**A, the second detection conductors **52**B, the first gate conductors **53**A and the second gate conductors **53**B. Of these conductors, the first detection conductors **52**A and the second detection conductor **52**B have the same configurations as those in the semiconductor device A**12** described before, so that the description is omitted.
(102) As shown in FIG. **23**, as viewed along the thickness direction z, each of the first gate conductors **53**A is in the form of a strip elongated in the first direction x. The width B**2***a* of the first gate conductors **53**A is smaller than the width Ba of the first leads **51**A. Each first gate conductor **53**A is made of a metal piece. As shown in FIG. **24**, the end of each first gate conductor **53**A on the first side in the first direction x is connected to the second electrode **42** of a respective first semiconductor element **40**A with a second bonding layer **49**. The end of each first gate conductor **53**A on the second side in the first direction x is connected to the first gate wiring layer **22**A with a first bonding layer **29**.
(103) As shown in FIG. **24**, each of the first gate conductors **53**A has a has a first layer **533** and a pair of second layers **534**. The first layer **533** is made of an alloy containing iron and nickel. Examples of the alloy are the same as those for the first layer **521**A of the pillow part **521** of the first detection conductor **52**A. The paired second layers **534** are made of a metal.
(104) Examples of the metal are the same as those for the paired second layers **521**B of the pillow part **521** of the first detection conductor **52**A. The first layer **533** is sandwiched between the paired second layers **534** in the thickness direction z. In this way, the first gate conductor **53**A is a laminate of a plurality of metal layers in the thickness direction z. The ratio of the thickness t**5***a* of the first layer **533** at the end on the first side in the first direction x and the thickness t**6***a* of the first layer **533** at the end on the second side in the first direction x may be t**5***a*: t**6***a*=1:2, for example. The coefficient of linear expansion of the first gate conductor **53**A having such a configuration is in a range of 0 to 8×10.sup.−6/° C. In contrast, the coefficient of linear expansion of the first conductive layer **20**A is about 16×10.sup.−6/° C. Thus, the coefficient of linear expansion of the first gate conductors **53**A is smaller than that of the first conductive layer **20**A. The first layer **533** of the first gate conductor **53**A has a transitional surface **533**A. The transitional surface **533**A is a curved surface located at the section where the thickness of the first layer **533** changes from the thickness t**5***a* to the thickness t**6***a*.
(105) As shown in FIG. **25**, as viewed along the thickness direction z, each of the second gate conductors **53**B is in the form of a strip elongated in the first direction x. The width B**2***b* of the second gate conductors **53**B is smaller than the width Bb of the second leads **51**B. Each second gate conductor **53**B is made of a metal piece. As shown in FIG. **26**, the end of each second gate conductor **53**B on the first side in the first direction x is connected to the second gate wiring layer **22**B with a first bonding layer **29**. The end of each second gate conductor **53**B on the second side in the first direction x is connected to the second electrode **42** of a respective second semiconductor element **40**B with a second bonding layer **49**.
(106) As shown in FIG. **26**, each of the second gate conductors **53**B has a first layer **533** and a pair of second layers **534**. The ratio of the thickness t**5***b* of the first layer **533** at the end on the first side in the first direction x and the thickness t**6***b* of the first layer **533** at the end on the second side in the first direction x may be t**5***b*: t**6***b*=2:1, for example. Other configurations of the first layer **533** and the second layers **534** of each second gate conductor **53**B are the same as those of the first layer **533** and the second layers **534** of each first gate conductor **53**A, so that the description of such configurations is omitted. Note that the coefficient of linear expansion of the second conductive layer **20**B is generally equal to that of the first conductive layer **20**A. Thus, the

coefficient of linear expansion of the second gate conductor **53**B is smaller than that of the second conductive layer **20**B. The first layer **533** of the second gate conductor **53**B has a transitional surface **533**A. The transitional surface **533**A is a curved surface located at the section where the thickness of the first layer **533** changes from the thickness t**5***b* to the thickness t**6***b*.
(107) The advantages of the semiconductor device A**10** are described below.
(108) The semiconductor device A**10** includes the first semiconductor elements **40**A each having a first electrode **41** and a second electrode **42** and bonded for electrical connection to the first conductive layer **20**A, the first leads **51**A connected to the first electrodes **41** and the second conductive layer **20**B, the first detection conductors **52**A, and the second detection conductors **52**B. The first detection conductors **52**A are connected to the first electrodes **41**. The first gate conductors **53**A are connected to the second electrodes **42**. In at least either of the first detection conductors **52**A and the first gate conductors **53**A, the ends connected to the first semiconductor elements **40**A have a coefficient of linear expansion smaller than that of the first conductive layer **20**A. Such an arrangement reduces at least either of the thermal stress generated between the first electrodes **41** and the first detection conductors **52**A and the thermal stress generated between the second electrodes **42** and the first gate conductors **53**A. As a result, the possibility of detachment from the first semiconductor elements **40**A is reduced in at least either of the first detection conductors **52**A and the first gate conductors **53**A, which assures the reliability of the semiconductor device A**10**.
(109) In the semiconductor device A**10**, the first detection conductors **52**A and the first gate conductors **53**A have pillow parts **521**, **531** connected to the first semiconductor elements **40**A and wire parts **522**, **532** connected to the pillow parts **521**, **531**. The coefficient of linear expansion of the pillow parts **521**, **531** is smaller than that of the first conductive layer **20**A. With such an arrangement, the first detection conductors **52**A and the first gate conductors **53**A have, at their ends connected to the first semiconductor elements **40**A, a coefficient of linear expansion smaller than the first conductive layer **20**A.
(110) Each pillow part **521**, **531** has a first layer **521**A, **531**A made of an alloy containing iron and nickel and a pair of second layers **521**B, **531**B made of a metal different from the first layer **521**A, **531**A. The first layer **521**A, **531**A is sandwiched between the paired second layers **521**B, **531**B in the thickness direction z. With such an arrangement, the coefficient of linear expansion of the pillow parts **521**, **531** can be made smaller than that of the first conductive layer **20**A. Also, such an arrangement allows for reliable connection of the pillow parts **521**, **531** to both of the first semiconductor elements **40**A and the wire parts **522**, **532**.
(111) The first layers **521**A, **531**A of the pillow parts **521**, **531** can be made of a semiconductor material, instead of an alloy containing iron and nickel. In such a case, the coefficient of linear expansion of the pillow parts **521**, **531** becomes closer to that of the first semiconductor element **40**A. This allows for more effective reduction of at least either of the thermal stress generated between the first electrodes **41** and the first detection conductors **52**A and the thermal stress generated between the second electrodes **42** and the first gate conductors **53**A.
(112) In the semiconductor device A**11**, the first layer **521**A, **531**A of each pillow part **521**, **531** has a lower layer portion **521**C, **531**C and an upper layer portion **521**D, **531**D. As viewed along the thickness direction z, the area of the upper layer portion **521**D, **531**D is larger than that of the lower layer portion **521**C, **531**C. With such an arrangement, the pillow part **521**, **531** has a relatively large area for connecting the wire part **522**, **532**, which makes easier the connection of the wire part **522**, **532** to the pillow part **521**, **531**.
(113) In the semiconductor device A**12**, each first detection conductor **52**A is made of a metal piece. The coefficient of linear expansion of the first detection conductors **52**A is smaller than that of the first conductive layer **20**A. With such an arrangement, the first detection conductors **52**A have, at their ends connected to the first semiconductor elements **40**A, a coefficient of linear expansion smaller than the first conductive layer **20**A.

(114) Each of the first detection conductors **52**A of the semiconductor device A**12** has a first layer **523** made of an alloy containing iron and nickel and a pair of second layers **524** made of a metal different from the first layer **523**. The first layer **523** is sandwiched between the paired second layers **524** in the thickness direction z. With such an arrangement, the coefficient of linear expansion of the first detection conductors **52**A can be made smaller than that of the first conductive layer **20**A. Also, such an arrangement allows for reliable connection of the first detection conductors **52**A to both of the first semiconductor elements **40**A and the first detection wiring layer **21**A.
(115) In the semiconductor device A**10**, the first detection wiring layer **21**A and the first gate wiring layer **22**A are formed on the front surface **101** of the insulating support member **10**. Such an insulating support member **10**, formed with the first conductive layer **20**A, the first detection wiring layer **21**A and the first gate wiring layer **22**A on the front surface **101**, can be easily made by using a DBC (trademark) substrate.
(116) The semiconductor device A**10** further includes the second semiconductor elements **40**B each having a first electrode **41** and a second electrode **42** and bonded for electrical connection to the second conductive layer **20**B, the second leads **51**B connected to the first electrodes **41** of the second semiconductor elements **40**B, the second detection conductors **52**B and the second gate conductors **53**B. The second detection conductors **52**B are connected to the first electrodes **41** of the second semiconductor elements **40**B. The second gate conductors **53**B are connected to the second electrodes **42** of the second semiconductor elements **40**B. In at least either of the second detection conductors **52**B and the second gate conductors **53**B, the ends connected to the second semiconductor elements **40**B have a coefficient of linear expansion smaller than that of the second conductive layer **20**B. Such an arrangement reduces at least either of the thermal stress generated between the first electrodes **41** of the second semiconductor elements **40**B and the second detection conductor **52**B and the thermal stress generated between the second electrodes **42** of the second semiconductor elements **40**B and the second gate conductors **53**B. As a result, the possibility of detachment from the second semiconductor elements **40**B is reduced in at least either of the second detection conductors **52**B and the second gate conductors **53**B.
(117) The semiconductor device A**10** further includes the first input terminal **31** and the second input terminal **32**. The first input terminal **31** is electrically connected to the first conductive layer **20**A. The second input terminal **32** is electrically connected to the second leads **51**B. The second leads **51**B are connected to the second input terminal **32**. Thus, the second input terminal **32** and the second leads **51**B can be formed as an integral part, which reduces the number of parts of the semiconductor device A**10**.
(118) The first input terminal **31** and the second input terminal **32** are located on the first side in the first direction x. The first input terminal **31** and the second input terminal **32** are spaced apart from each other in the thickness direction z. As viewed along the thickness direction z, part of the second input terminal **32** (the second terminal part **322**) overlaps with the first input terminal **31**. With such an arrangement, during the use of the semiconductor device A**10**, the magnetic field generated from the second input terminal **32** reduces the inductance of the first input terminal **31**.
Second Embodiment
(119) A semiconductor device A**20** according to a second embodiment is described below based on FIGS. **27-32**. In these figures, the elements that are identical or similar to those of the semiconductor device A**10** are denoted by the same reference signs as those used for the semiconductor device A**10** and are not described. In FIG. **27**, the sealing resin **60** is illustrated as transparent for the convenience of understanding. Thus, the sealing resin **60** are indicated by imaginary lines (two-dot chain lines).
(120) The semiconductor device A**20** differs from the semiconductor device A**10** in that the semiconductor device A**20** is provided with a pair of insulating layers **23** and does not include a metal substrate **69**. The semiconductor device A**20** differs from the semiconductor device A**10** also

in configurations of the first detection wiring layer **21**A, the second detection wiring layer **21**B, the first gate wiring layer **22**A, the second gate wiring layer **22**B, the first semiconductor elements **40**A, the second semiconductor elements **40**B, the first leads **51**A and the second leads **51**B.

(121) As shown in FIGS. **27**, **29** and **30**, the paired insulating layers **23** are arranged on the first conductive layer **20**A and the second conductive layer **20**B. The insulating layers **23** are spaced apart from each other in the first direction x. Each of the insulating layers **23** is in the form of a strip elongated in the second direction y. The insulating layer **23** that is offset toward the first side in the first direction x is on the first conductive layer **20**A. The insulating layer **23** that is offset toward the second side in the first direction x is on the second conductive layer **20**B. The insulating layers **23** may be made of glass epoxy resin, for example.

(122) As shown in FIGS. **27**, **29** and **30**, the first detection wiring layer **21**A and the first gate wiring layer **22**A are arranged on one of the insulating layers **23** that is on the first conductive layer **20**A. The second detection wiring layer **21**B and the second gate wiring layer **22**B are arranged on the other one of the insulating layers **23** that is on the second conductive layer **20**B. Thus, in the semiconductor device A**20** again, as viewed along the thickness direction z, the first detection wiring layer **21**A, the second detection wiring layer **21**B, the first gate wiring layer **22**A and the second gate wiring layer **22**B overlap with the front surface **101** of the insulating support member **10**.

(123) As shown in FIGS. **27** and **29**, the first semiconductor elements **40**A are offset toward the second side in the first direction x from the one of the insulating layers **23** that is on the first conductive layer **20**A. In each of the first semiconductor elements **40**A, the second electrode **42** is offset toward the first side in the first direction x.

(124) As shown in FIGS. **27** and **30**, the second semiconductor elements **40**B are offset toward the first side in the first direction x from the other one of the insulating layers **23** that is on the second conductive layer **20**B. In each of the second semiconductor elements **40**B, the second electrode **42** is offset toward the second side in the first direction x.

(125) As shown in FIG. **27**, the dimension of each first lead **51**A in the first direction x is smaller than that in the semiconductor device A**10**. Also, the dimension of each second lead **51**B in the first direction x is smaller than that in the semiconductor device A**10**.

(126) As shown in FIGS. **27** and **28**, a pair of detection terminals **34** takes the place of the pair of gate terminals **35** of the semiconductor device A**10**. Also, a pair of gate terminals **35** takes the place of the pair of detection terminal **34** of the semiconductor device A**10**. As shown in FIG. **28**, the back surface **102** of the insulating support member **10** is exposed from the bottom surface **62** of the sealing resin **60**.

(127) As shown in FIG. **31**, as viewed along the thickness direction z, the first detection conductors **52**A and the first gate conductors **53**A extend from the first semiconductor elements **40**A toward the first side in the first direction x. As shown in FIG. **32**, as viewed along the thickness direction z, the second detection conductors **52**B and the second gate conductors **53**B extend from the second semiconductor elements **40**B toward the second side in the first direction x. Other configurations of each of the first detection conductors **52**A, the second detection conductors **52**B, the first gate conductors **53**A and the second gate conductors **53**B are the same as those in the semiconductor device A**10**, so that the description of such configurations is omitted.

First Variation of the Second Embodiment

(128) A semiconductor device A**21**, which is the first variation of the semiconductor device A**20**, is described below based on FIGS. **33** and **34**. The semiconductor device A**21** differs from the semiconductor device A**20** in configuration of the first detection conductors **52**A and the second detection conductors **52**B.

(129) As shown in FIG. **33**, as viewed along the thickness direction z, each of the first detection conductors **52**A is in the form of a strip extending from the first semiconductor element **40**A toward the first side in the first direction x. The width B**1***a* of the first detection conductors **52**A is

smaller than the width Ba of the first leads **51**A. Each first detection conductors **52**A is made of a metal piece. Other configurations of the first detection conductors **52**A are the same as those of the first detection conductors **52**A of the semiconductor device A**12**, so that the description is omitted.
(130) As shown in FIG. **34**, as viewed along the thickness direction z, each of the second detection conductors **52**B is in the form of a strip extending from the second semiconductor element **40**B toward the second side in the first direction x. The width B**1***b* of the second detection conductors **52**B is smaller than the width Bb of the second leads **51**B. Each second detection conductor **52**B is made of a metal piece. Other configurations of the second detection conductors **52**B are the same as those of the second detection conductors **52**B of the semiconductor device A**12**, so that the description is omitted.
Second Variation of the Second Embodiment
(131) A semiconductor device A**22**, which is the second variation of the semiconductor device A**20**, is described below based on FIGS. **35** and **36**. The semiconductor device A**22** differs from the semiconductor device A**20** in configuration of the first detection conductors **52**A, the second detection conductors **52**B, the first gate conductors **53**A and the second gate conductors **53**B. Of these conductors, the first detection conductors **52**A and the second detection conductors **52**B have the same configurations as those of the semiconductor device A**21**, so that the description is omitted.
(132) As shown in FIG. **35**, as viewed along the thickness direction z, each of the first gate conductors **53**A is in the form of a strip extending from the first semiconductor element **40**A toward the first side in the first direction x. The width B**2***a* of the first gate conductors **53**A is smaller than the width Ba of the first leads **51**A. The length L**2***a* of the first gate conductors **53**A is smaller than the length L**1***a* of the first detection conductors **52**A. Each first gate conductor **53**A is made of a metal piece. Other configurations of the first gate conductors **53**A are the same as those of the first gate conductors **53**A of the semiconductor device A**13**, the description is omitted.
(133) As shown in FIG. **36**, as viewed along the thickness direction z, each of the second gate conductors **53**B is in the form of a strip extending from the second semiconductor element **40**B toward the second side in the first direction x. The width B**2***b* of the second gate conductors **53**B is smaller than the width Bb of the second leads **51**B. The length L**2***b* of the second gate conductors **53**B is smaller than the length L**1***b* of the second detection conductors **52**B. Each second gate conductor **53**B is made of a metal piece. Other configurations of the second gate conductors **53**B are the same as those of the second gate conductors **53**B of the semiconductor device A**13**, so that the description is omitted.
(134) The advantages of the semiconductor device A**20** are described below.
(135) As with the semiconductor device A**10**, the semiconductor device A**20** includes the first semiconductor elements **40**A each having a first electrode **41** and a second electrode **42** and bonded for electrical connection to the first conductive layer **20**A, the first leads **51**A connected to the first electrodes **41** and the second conductive layer **20**B, the first detection conductors **52**A, and the second detection conductors **52**B. The first detection conductors **52**A are connected to the first electrodes **41**. The first gate conductors **53**A are connected to the second electrodes **42**. In at least either of the first detection conductors **52**A and the first gate conductors **53**A, the ends connected to the first semiconductor elements **40**A have a coefficient of linear expansion smaller than that of the first conductive layer **20**A, which assures the reliability of the semiconductor device A**20**.
(136) The semiconductor device A**20** is provided with an insulating layer **23** on the first conductive layer **20**A. The first detection wiring layer **21**A and the first gate wiring layer **22**A are arranged on the insulating layer **23**. Such an arrangement makes it possible to increase the area of the first conductive layer **20**A as viewed along the thickness direction z and thereby to promote heat dissipation from the semiconductor device A**20**. Moreover, by arranging the first semiconductor elements **40**A offset from the insulating layer **23** toward the second side in the first direction x, the dimension of each first lead **51**A in the first direction x can be reduced. As a result, the parasitic

resistance of the semiconductor device A**20** can be reduced.

(137) The semiconductor device according to the present disclosure is not limited to the foregoing embodiments. The specific configuration of each part of the semiconductor device may be varied in design in many ways.

Claims

1. A semiconductor device comprising: an insulating support member having a front surface; a first conductive layer disposed on the front surface; a first semiconductor element having a first side facing the front surface and a second side facing away from the first side in a thickness direction of the insulating support member, the first semiconductor element being provided with a first electrode and a second electrode on the second side and with a third electrode on the first side, the third electrode being disposed on and electrically bonded to the first conductive layer, the first semiconductor element being formed of a semiconductor material mainly composed of silicon carbide; a first lead connected to the first electrode, the first lead being made of copper or a copper alloy and having a shape of a strip; a first gate conductor having one end connected to the second electrode; a first bonding layer disposed between the first lead and the first electrode; a second bonding layer disposed between the first gate conductor and the second electrode; and a second conductive layer spaced apart from the first conductive layer, wherein the second bonding layer is formed as a layer different from the first bonding layer, and the second bonding layer has a smaller coefficient of linear expansion than that of the first conductive layer or that of the second conductive layer.
2. The semiconductor device according to claim 1, further comprising a first sense conductor having one end connected to the first electrode.
3. The semiconductor device according to claim 2, further comprising a third bonding layer disposed between the first sense conductor and the first electrode.
4. The semiconductor device according to claim 3, wherein the third bonding layer is formed as a layer different from the first bonding layer.
5. The semiconductor device according to claim 4, wherein the second bonding layer and the third bonding layer are each formed of at least one metal material different from a metal material for forming the first bonding layer.
6. The semiconductor device according to claim 2, wherein the first sense conductor comprises a wire or a metal piece.
7. The semiconductor device according to claim 2, wherein the first gate conductor and the first sense conductor are each narrower than the first lead.
8. The semiconductor device according to claim 1, wherein the insulating support member contains a ceramic material.
9. The semiconductor device according to claim 1, wherein the first lead is connected to the second conductive layer.
10. The semiconductor device according to claim 1, further comprising a second semiconductor element formed of a semiconductor material mainly composed of silicon carbide, wherein the second semiconductor element is provided with at least one electrode disposed on and electrically bonded to the second conductive layer.
11. The semiconductor device according to claim 1, wherein the first gate conductor comprises a wire or a metal piece.
12. The semiconductor device according to claim 1, wherein the first bonding layer is made of a solder or a baked silver.
13. The semiconductor device according to claim 1, wherein the second bonding layer contains an alloy containing iron and nickel.
14. The semiconductor device according to claim 1, wherein the second bonding layer contains one

of a metal, copper, a copper alloy, aluminum and an aluminum alloy.

15. A semiconductor device comprising: an insulating support member having a front surface; a first conductive layer disposed on the front surface; a first semiconductor element having a first side facing the front surface and a second side facing away from the first side in a thickness direction of the insulating support member, the first semiconductor element being provided with a first electrode and a second electrode on the second side and with a third electrode on the first side, the third electrode being disposed on and electrically bonded to the first conductive layer, the first semiconductor element being formed of a semiconductor material mainly composed of silicon carbide; a first lead connected to the first electrode, the first lead being made of copper or a copper alloy and having a shape of a strip; a first gate conductor having one end connected to the second electrode; a first bonding layer disposed between the first lead and the first electrode; a second bonding layer disposed between the first gate conductor and the second electrode; a first sense conductor having one end connected to the first electrode; and a second conductive layer spaced apart from the first conductive layer, wherein the second bonding layer is formed as a layer different from the first bonding layer, and the first sense conductor and the first gate conductor each have a smaller coefficient of linear expansion than that of the first conductive layer or that of the second conductive layer.

16. A semiconductor device comprising: an insulating support member having a front surface; a first conductive layer disposed on the front surface; a first semiconductor element having a first side facing the front surface and a second side facing away from the first side in a thickness direction of the insulating support member, the first semiconductor element being provided with a first electrode and a second electrode on the second side and with a third electrode on the first side, the third electrode being disposed on and electrically bonded to the first conductive layer, the first semiconductor element being formed of a semiconductor material mainly composed of silicon carbide; a first lead connected to the first electrode, the first lead being made of copper or a copper alloy and having a shape of a strip; a first gate conductor having one end connected to the second electrode; a first bonding layer disposed between the first lead and the first electrode; a second bonding layer disposed between the first gate conductor and the second electrode; a first sense conductor having one end connected to the first electrode; a first sense wire connected to the first sense conductor; and a first gate wiring layer connected to the first gate conductor, wherein the second bonding layer is formed as a layer different from the first bonding layer, and the first sense wire and the first gate wiring layer overlap with the front surface as viewed in the thickness direction.

17. The semiconductor device according to claim 16, wherein the first sense wire and the first gate wiring layer are disposed on the front surface.

18. The semiconductor device according to claim 16, further comprising an insulating layer disposed on the first conductive layer, wherein the first sense wire and the first gate wiring layer are disposed on the insulating layer.